LIBBY — PRECLINICAL HORIZON

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. Reversing post-transplant MHC class II downregulation to restore graft-versus-leukemia (conceptual strategy)

Target: HLA-restricted GVL / post-HCT immune escape **Stage:** Early translational **Evidence strength:** Emerging

The mechanism that best matches this patient's late GVL-escape relapse, testable with drugs already on the shelf, but gated on first learning whether the escape is epigenetic or genetic, and useless against the immediate blast burden.

This reads like a textbook late GVL-escape relapse: about six years in remission after a matched-sibling transplant, then breakthrough, and this patient's own flow already flags HLA-DR as variably decreased. That is the exact escape route the supporting work describes, and two independent post-HCT relapse cohorts (DiPersio and Vago groups) showed the class II loss was epigenetic and reversible with interferon-gamma rather than genetic. If that is what is happening here, an interferon-gamma or hypomethylating priming step could make a second transplant, DLI, or WT1/HA-directed TCR-T actually bite. What holds it at emerging rather than higher: nobody has yet characterized whether this specific relapse is epigenetic silencing or genetic HLA loss, and priming does nothing for the second case. It is an adjunct that potentiates the cellular strategies, not a therapy that reduces the current 85% blast burden.

Mechanism. After allo-HCT, relapsing AML blasts often switch off HLA class II epigenetically instead of mutating it away, which hides them from donor CD4 T cells. The silencing is reversible: interferon-gamma or a hypomethylating agent can turn MHC-II back on, re-exposing the blasts to donor T cells and to any HLA-restricted TCR-T given as consolidation.

Translatability. Moderate: strong human paired-sample data for the escape biology, but the priming-to-GVL step has not been run as a protocol, and this patient's escape type is unconfirmed.

Developability. Closest to reach of the set: interferon-gamma and azacitidine/decitabine are already available, so a re-induction step could be tested without new chemistry. No protocol packages it as GVL restoration yet.

Counter-productive MoA. Moderate (Turning class II back on and inflaming the graft can drive GVHD, the same donor-T-cell activity being harnessed for GVL.)

Key risks.

- Re-inducing HLA class II and inflaming the graft could aggravate GVHD.
- If the escape is genetic HLA loss (6p uniparental disomy) rather than epigenetic silencing, priming does nothing and a donor change or non-HLA-restricted route is needed.
- Adjunctive only: it sharpens the cellular therapies and does not touch the current 85% blast burden.

Open questions.

- Is this relapse epigenetic MHC-II silencing (priming-responsive) or genetic HLA loss (priming-refractory)?
- Would an interferon-gamma or hypomethylating prime measurably restore blast antigen presentation before a second transplant or TCR-T consolidation?

Key references. pmid:30911134; pmid:30380364; pmid:19641204

2. Epitope-edited donor HSPC graft enabling CD123/CD33-directed therapy without hematopoietic toxicity (cell therapy)

Target: CD123+/CD33+ blasts (antigen-escape / on-target-off-leukemia hedge) **Stage:** In vivo (animal)
Evidence strength: Emerging

The candidate that fits the planned second transplant most cleanly and has the most mature preclinical proof, but a large manufacturing and safety lift that offers nothing for the immediate disease burden and rests on unquantified antigen density.

This patient is CD123+ and CD33+ by flow with documented clonal evolution, and a second allo-HCT is already on the table, so the graft is genuinely something that could be engineered. It also has the most mature evidence in the set: the Genovese group showed in mouse models that base-edited donor HSPCs resist CAR-T and antibodies while staying functional, which widens the window for AML-directed immunotherapy after transplant. Two things keep it at rank 2 rather than 1. First, it layers base-editing genotoxicity and an edited-cell regulatory package onto an already high-risk second transplant, and the manufacturing and time cost buys nothing against a marrow that is currently about 85% blasts. Second, the CD123 and CD33 positivity here is qualitative flow only; antigen density has not been quantified, so how hard a targeting agent could actually run is still unknown. The concept fits the planned HCT2 more neatly than anything else, but it is nowhere near clinical readiness.

Mechanism. Base editing changes a few residues in the CD123 (or CD33/FLT3/KIT) epitope on donor stem cells so antibodies and CAR-T no longer see the healthy graft, while the protein keeps binding its ligand and working. The engrafted, edited marrow becomes invisible to the targeting agent, so a CD123- or CD33-directed drug can be pushed against leukemia without wiping out normal hematopoiesis.

Translatability. Moderate: real in-vivo proof-of-concept in edited-HSPC xenografts, though CD123/CD33 positivity here is qualitative only, so antigen density (which sets the achievable dose intensity) is unmeasured.

Developability. Far from reach: needs an edited donor graft manufactured on top of a complex second transplant, plus a regulatory package for a base-edited cell product. The preclinical proof is real, but nothing is enrolling.

Counter-productive MoA. Low (Editing shields normal marrow rather than the tumor, so little direct pathway conflict; the risk sits in genotoxicity and the delivered agent's toxicity.)

Key risks.

- Adds base-editing genotoxicity and off-target unknowns to a high-risk second transplant.
- Manufacturing and time cost do nothing for a marrow that is about 85% blasts and needs cytoreduction now.
- Benefit still depends on then giving a CD123/CD33 agent hard enough to matter, which reintroduces that agent's own toxicity and rests on antigen density that has not been quantified.

Open questions.

- What are the quantitative CD123 and CD33 antigen densities on these blasts, and are they high enough to justify the edited-graft approach?
- Can an edited donor graft be manufactured and qualified within the timeline a second transplant for this patient would allow?

Key references. pmid:37648862

3. TCR-T against next-generation hematopoietic minor-H antigens (UTA2-1 and other A*02:01-restricted mHAgs beyond HA-1/HA-2) (cell therapy)

Target: HLA-restricted GVL-without-GVHD immunotherapy handles **Stage:** Early translational **Evidence strength:** Emerging

A genuine fallback antigen that keeps the GVL-without-GVHD route alive if HA-1/HA-2 comes back uninformative, but still an antigen-discovery finding with no product and the same double genotype gate.

The primary reason this case came to Libby is the HLA-restricted GVL route, and the clinical TCR-T products cover only HA-1 and HA-2, with eligibility hanging on the sibling pair carrying that exact disparity. UTA2-1 and the wider set of A*02:01 hematopoietic minor-H antigens give a fallback antigen if the HA-1/HA-2 genotyping comes back uninformative, which keeps the GVL-without-GVHD door open. What holds it at rank 3: the supporting evidence is a single 2013 antigen-discovery report (Mutis group) built on patient CTL clones, with no engineered receptor, no clinical-grade product, and no trial. It is doubly gated, on the still-unresolved A*02:01 status and on the patient and sibling actually differing at the antigen, the same eligibility trap that limits HA-1/HA-2. The delivery platform (donor-derived TCR-T after allo-HCT) is at least the same one already charted for HA-1/HA-2.

Mechanism. UTA2-1 is an HLA-A*02:01-restricted peptide from a biallelic, blood-lineage-restricted gene. A donor T cell that recognizes the recipient's version of the peptide kills hematopoietic cells, including leukemia, while sparing the epithelial tissues that drive GVHD. It broadens the usable mismatch pool beyond HA-1 and HA-2 to donor/recipient pairs that happen to differ at this locus.

Translatability. Moderate: a validated hematopoiesis-restricted antigen with real patient-CTL cytotoxicity data, but no engineered TCR, and it is gated on unresolved A*02:01 plus a sibling disparity that may not exist.

Developability. Far from reach: no clinical-grade TCR against UTA2-1 exists, so a receptor would have to be isolated, validated, and built into a product. The donor-TCR-T-after-HCT platform is at least the one already used for HA-1/HA-2.

Counter-productive MoA. Low (On-target reach is hematopoiesis-restricted, so off-leukemia epithelial GVHD is unlikely; the main failure mode is antigen-loss escape, not pathway conflict.)

Key risks.

- Needs the specific recipient-positive / donor-negative UTA2-1 disparity, which may be no more present in this sibling pair than the HA-1/HA-2 one.
- On-target reach is limited to hematopoietic reconstitution, and antigen-loss escape stays possible given this patient's clonal evolution.
- Gated on A*02:01, which is still unresolved.

Open questions.

- Do the patient and sibling donor differ at UTA2-1 in the direction that enables GVL, and is the patient A*02:01-positive?
- Can a clinical-grade TCR against UTA2-1 be isolated and built onto the existing donor-TCR-T platform?

Key references. pmid:23079962

4. Next-generation TP53 Y220C stabilizer chemotypes (covalent tyrosine-mimic fragments; novel covalent scaffolds) (small molecule)

Target: TP53 Y220C aberration (reactivation handle) **Stage:** In vitro **Evidence strength:** Speculative

A structurally distinct covalent backup to rezatapopt worth watching, but tool fragments with no cellular kill, doubly gated on a TP53/Y220C status that is still pending, and probably a combination agent at best.

If TP53 is confirmed and the variant turns out to be Y220C, rezatapopt is the clinical option, but it is a single molecule from one sponsor and its single-agent myeloid activity is blunted by MDM2, XPO1, and BCL-2 feedback. A structurally distinct covalent backup chemotype would matter if rezatapopt fails, is inaccessible, or needs a longer-residence binder. That is the whole reason it is logged, and it sits at rank 4 because the case is thin in three ways. The published matter is fragments and virtual-screen hits (Boeckler and Li groups) with thermal-stability and crystallography data but no cellular kill or apoptosis, so cellular potency, PK, and a myeloid model would all have to come first. It is doubly gated: confirm the TP53 aberration, then confirm it is Y220C, and a non-Y220C lesion makes the entire chemotype irrelevant. And the Y220C aberration itself is unconfirmed here, held as pending on a limited relapse panel, so this is a hedge behind a hedge rather than a plan.

Mechanism. The Y220C substitution opens a surface crevice that destabilizes the p53 DNA-binding domain. These compounds dock into that crevice, several forming a covalent bond to Cys220, and raise the mutant's melting temperature so it holds a wild-type-like fold. The covalent tyrosine-mimic fragments literally put back the aromatic ring the mutation removed.

Translatability. Low: recombinant thermal-stability and X-ray data only, no cellular apoptosis, and the whole chemotype depends on first confirming an as-yet-unconfirmed Y220C lesion.

Developability. Far from reach: fragments and virtual-screen hits, not lead compounds. Cellular potency, apoptosis, PK, and a myeloid model would all be needed before any dosing, leaving it years behind rezatapopt.

Counter-productive MoA. Low (Refolding Y220C is not directly counter-productive, but single-agent reactivation is weakly pro-apoptotic in myeloid cells and may underperform without a partner.)

Key risks.

- No cellular kill shown; the myeloid Y220C data predict reactivation alone is weakly pro-apoptotic and would need a venetoclax or MDM2/XPO1 partner.
- Doubly gated: confirm the TP53 aberration, then confirm Y220C; a non-Y220C lesion makes the chemotype irrelevant, and the aberration itself is pending here.
- Likely superseded by rezatapopt if that program succeeds in myeloid disease.

Open questions.

- Once TP53 status resolves, is the lesion actually Y220C?

- Can any of these fragments be optimized to a cell-active lead with apoptosis in a myeloid Y220C model, alone or with venetoclax?

Key references. pmid:39698266; pmid:39946082

5. Mutant-p53 destabilization via DNAJA1/HSP40 inhibition or statin repurposing (repurposed drug)

Target: TP53 Y220C aberration (destabilization handle) **Stage:** In vitro **Evidence strength:** Speculative

A cheap, repurposable degradation hypothesis worth probing, but every study is cross-tumor, the myeloid biology is untested, and clearing mutant p53 without restoring suppressor function may accomplish little.

This is the opposite bet to reactivation: rather than refold Y220C, degrade it. It is worth flagging because statins are cheap, widely taken, and easy to probe for a repurposing signal, and the approach does not depend on reactivator pharmacology. It ranks last for reasons that stack up. Everything published (Iwakuma group) is in solid-tumor conformational mutants, not AML, so the case here is cross-tumor only, with no myeloid dose-response and DNAJA1 inhibitors still just tool compounds. There is also a mechanism-level catch: degrading mutant p53 strips out any gain-of-function but does not restore tumor-suppressor activity, so in a clone that already behaves as p53-null it may do very little on its own. For this patient it is a mechanism to probe, not something to prescribe, and it sits philosophically at odds with the rezatapopt reactivation route, so the two should not be assumed to combine.

Mechanism. DNAJA1, an HSP40 co-chaperone, binds misfolded conformational mutant p53 and shields it from CHIP-mediated degradation. Blocking DNAJA1, or lowering mevalonate output with a statin (which impairs the DNAJA1 interaction), shunts the mutant protein to the proteasome. Y220C is a thermolabile, misfolding mutant, so it is a plausible DNAJA1 client.

Translatability. Low: all data are in solid-tumor conformational mutants, never AML Y220C, so whether myeloid blasts behave as DNAJA1 clients and respond to statin-driven degradation is untested.

Developability. Mixed: statins are approved and could be tested as a repurposing signal quickly, but there is no AML program, no myeloid dose-response, and DNAJA1 inhibitors remain tool compounds.

Counter-productive MoA. Moderate (Clearing mutant p53 without restoring suppressor function can leave a p53-null-behaving clone unchanged, so the intended benefit may not materialize.)

Key risks.

- Degrading mutant p53 removes gain-of-function but does not restore suppressor activity, so it may do little in a clone that already behaves as p53-null.
- Whether AML Y220C is a DNAJA1 client that responds to statin-driven degradation is untested.
- Single-agent antileukemic effect of statins is unproven; this is a mechanism to probe, not to prescribe.

Open questions.

- Does AML Y220C behave as a DNAJA1 client and degrade under statin exposure?
- Would degradation alone change leukemic behavior in a clone that already functions as p53-null?

Key references. pmid:27775703